

1) Which material type is more commonly injection moulded, thermoplastics or thermosets? Can you name some thermoplastics and thermosets? What is an important characteristic of thermoplastics and of thermosets.

Thermoplastics are more commonly used in injection moulding.

Common thermoplastics: PE, PP, PLA, PET, ABS, PVC...

Common thermosets: Epoxy, silicone, polyurethane.

An important property of thermoplastics is that they have a melting temperature, above which they are a liquid.

An important property of thermosets is that they transform from a liquid resin into a solid material with an increase in temperature by chemical crosslinking.

2) Some types of plastics are readily recycled if they can be repeatedly heated (melted) and cooled with little material degradation. Will the choice of HDPE for the O-frame allow for the part to be recyclable?

Yes, HDPE is a thermoplastic and is hence recyclable, but it may not be recyclable into the same product.

3) What machine settings have important effects on the part properties in injection moulding?

Machine settings:

- Injection pressure
- Holding pressure
- Temperature of the injector
- Temperature of the mould
- Temperature of the barrel

Everything that relates to pressure and temperature may affect the viscosity of the polymer and its ability to flow into the mould. The pressure also controls how the polymer chains are compacted into the die.

4) What 2 mechanisms provide heat to melt the polymer in the moulding machine barrel?

The 2 mechanisms that provide heat to the polymer are heat and mechanical action through friction, shear and pressure.

5) Why is the D-frame part appropriate for the process of injection moulding?

The D-frame part is suitable for injection moulding due to:

- Its geometry, as injection moulding can be used to produce complex and intricate pieces.
- Its size, as injection moulding can be used to produce small pieces.
- The desired quantity being large, as injection moulding can produce large quantities in a short amount of time.

6) A 3-plate mould for injection moulding is more complex and expensive than a 2-plate mould.

What advantage of the 3-plate system justifies this extra investment.

A 3-plate mould uses 3 plates to separate the parts from the sprue and the runner when the mould opens. The advantages over the 2-plate mould are:

- As the mould opens, the runner and the parts disconnect and drop into two containers placed under the mould.

- It allows automatic operation of the moulding machine, saving time and money during production.

7) How could the use of a 3-plate mould possibly correct the problem of weld lines on the part without needing to increase temperature?

A weld line forms when polymer flows around a cone or other convex section and meets from the opposite direction.

A 3-plate mould can avoid the partition and the formation of weld lines by allowing more flexibility in the positioning of the gates by the process designer. Even though the physical design of the O-frame and its mould layout is not given, it is possible that the gates could be moved to a more central location that would more evenly distribute plastic to all sides of the part, thus preventing weld lines.

8) Although the O-frame parts are successfully produced, it is eventually discovered that the parts from the initial run are considerably larger than designed.

Why do you think the parts come out too large from the initial run?

The parts are most likely larger than expected because the injection temperature and pressure are higher than planned. Both of these cause more plastic to be packed into the mould cavity and reduce the amount of shrinkage as the part cools.

a) With the shrinkage values in the table below for the same injection conditions, what should be the corrected diameter of the mould to obtain a piece of 112.5 mm diameter?

What to do if we cannot change the mould?

Plastic	Shrinkage
ABS	0.006
Nylon-6,6	0.020
Polycarbonate	0.007
Polyethylene	0.025
Polystyrene	0.004
PVC	0.005

0.025 is the shrinkage for polyethylene.

$$\begin{aligned}D_c &= D_p + D_p S + D_p S^2 \\&= D_p (1 + S + S^2) \\&= 112.5 (1 + 0.025 + 0.025^2) \\&= 115.3828125 \approx 115.383 \text{ mm}\end{aligned}$$

If the mould cannot be changed, we can adjust the temperatures, pressures

and compaction time to change the actual shrinkage rate. However, changing the process parameters might change the density and other properties of the final part or lead to the formation of other defects.